The Rising Sun: Shopping List

For the first prototype, the light only. Written 2026-10-18 for Dr. O.

No prices are listed because I cannot check them from here. Nothing below has been tried on real parts. If a listing does not match what is written, do not buy it, and ask first.

Buy for the first prototype

- Controller board: you already have the ESP32-S3 N16R8 from the Holopod build. Use one of those. Nothing to buy.
- LED strip: search "WS2812B LED strip 5V". It must say 5V, WS2812B (or WS2812), and "individually addressable". Buy 1 meter with 60 LEDs per meter. Black or white backing both work. Do NOT buy 12V strips, or strips that say "RGB" with only 4 wires and no data line.
- Power supply: a 5V supply rated 4 amps or more, with a screw terminal or a barrel plug and a matching adapter. 60 LEDs can draw up to 3.6 amps at full white. Do NOT use a phone charger.
- Resistor: 330 ohm, any small assortment kit has it.
- Capacitor: 1000 microfarad, 6.3V or higher. Also in most kits.
- Jumper wires: male to female and male to male, a small assortment.
- Breadboard or a few screw terminal blocks, so nothing has to be soldered for the first try.
- USB cable: it must carry data, not just power. If the board does not show up on the computer, a charge-only cable is the usual reason.

Buy only if the first light flickers

- A level shifter, 74AHCT125 (a small board, sold as "logic level shifter"). The board sends a 3.3V signal and some strips want 5V.

Buy for sound: no soldering

Dr. O and David did not enjoy soldering the tiny holes on the Astra-9 amplifiers, so the sound parts below are chosen to need none, or as little as possible. Buy them with the first order. They are wired after David has seen the light working, and the program that plays her voice and music is not written yet. Nothing below has been checked against a build, so check each listing before buying. If you are unsure about one, send Claude a screenshot of the listing and ask.

First choice: one all-in-one audio board

One ESP32-S3 board that already has the amplifier, the microphone and a connector for a speaker on it. It replaces the separate sun board, the amplifier boards and the microphone board, so there is nothing tiny to solder. Search for: "ESP32-S3 audio development board speaker amplifier microphone". Check the listing says all of these:

- ESP32-S3 (not plain ESP32, and not ESP32-S2). The S3 has the memory and speed the voice needs.
- A built-in speaker amplifier and a speaker connector (a small plug or screw terminal). Note the plug type. If it is not the same small JST plug as the Astra-9 speakers, either cut the plug off the speaker leads and use screw terminals, or buy the matching plug.
- A built-in microphone, or a microphone connector.
- Header pins that are already soldered on, with at least one spare general-purpose pin free for the LED strip data wire. A sealed box with no free pins will not work, because the LED strip needs one pin.
- USB-C, so it plugs into the computer the same way as your other boards.

Speakers: the same ones as Astra-9. Use "Buy it again" on your Amazon order of September 15, 2026 (3 watt, 8 ohm mini speakers). The amplifier modules in that order are not needed for this board.

Second choice, if no all-in-one board fits: separate boards that need no soldering

- Two I2S amplifier boards, MAX98357A, one per speaker. In the listing photos the header pins must already be attached, and the speaker output should be a screw terminal. Skip any listing that shows loose pins in a bag.
- One I2S microphone, INMP441, with the header pins already attached. Same rule.
- Jumper wires with plugs on both ends (male to female and male to male), so every connection is pushed in and nothing is soldered.
- The small green screw terminal blocks that held up best on Astra-9, for the speaker wires.

For either choice

- Support every wire so nothing hangs from a joint. Thin jumper wires came loose on Astra-9.
- Keep the speaker wires and amplifiers away from the LED strip power wires, which carry far more current than the Astra-9 eye lights. Astra-9 had a buzz that followed the eye brightness. It may happen here, and the cause there is not fully known.

- The ESP32-S3 has Wi-Fi and low energy Bluetooth but not the regular Bluetooth that phone speakers use. The sun cannot be a Bluetooth speaker. Music comes through Wi-Fi from a service such as Apple Music or Spotify.

Do not buy yet

- The full length of strip for the acrylic. How many LEDs it needs depends on the acrylic shape, which is not decided. Buy it after David has seen the prototype and you have picked the layout.
- The bigger power supply. It is sized from the final LED count: about 60 milliamps per LED at full white, and a safe supply has some room above that.
- The sound program. It is not written yet. The parts for it are listed above.

Before David starts

- Lay the parts out and tick them off against this list.
- Give him the First Prototype sheet. It lists every step.
- He does not need any account, login or key for the first prototype.
- Ask him to send you a photo of the strip lit. That is the only report you need today.